

Suresh Nair, director, sales and marketing, Leaptch Corporation, feels that SMT machines' sale is spread across all segments, but the sectors which are using more machines are lighting and automotive. "In 2010, we sold Fuji machines in telecom, lighting, automotive, specialised electronics and EMS industry segments," adds S.P. Arora, senior business manager, NMTronics.

According to N. Chandramohan, country head (SMT division), Juki India, the Indian SMT market buys about 350 machines, which includes both new and second-hand machines. However, Arora says, "It's difficult to quote, as there is no independent data available for the industry."

The year 2010 was a better year for the SMT machine suppliers as 50 per cent more machines were sold as compared to the recession-hit 2009. Going by industry estimates, the growth rate of the SMT equipment market was around 30 per cent, as many of the equipment purchase decisions which were on hold during 2009 matured in 2010. "So 2010 was an extraordinary year for most players from business perspective because all the shortfalls of business in 2009 got converted in 2010," explains Pradeep Tandon, country manager, Samsung.

However, Chandramohan believes that although the SMT equipment market has started moving forward, it has not taken off as expected. "The demand drivers like cellphones, set-top boxes (STBs) and other consumer electronics products are still dependant on China for manufacturing. There seems some revival though in the automotive sector," he adds.

Market forecasts

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confidence of investors, OEMs and other electronic assemblers are lining up plans for modernising capital equipment, capacity expansion and technological upgradation, spelling promising opportunities for the growth of the SMT sector.

With the launch of the new manufacturing policy, the Indian manufacturing segment will show tremendous potential, and industry reports indicate its steady growth will continue. This would, in turn, bring the market growth for electronics assembly products in India. A Frost & Sullivan report on Indian SMT market shows that the market earned revenues of Rs 2689.8 million in 2008 and estimates this to reach Rs 4415.1 million in 2012, growing at a CAGR of 13.2 per cent.

According to Frost & Sullivan research analyst K. Bharath, "With an increase in the perks awarded to foreign direct investments (FDI) by the government of India, many of the global companies are anticipated to set up their manufacturing facilities in the country, thereby augmenting the demand for SMT equipment."

"I believe the next five years will be excellent for the SMT market in India, which has started to grow in real terms," adds Suresh Nair.

However, Dilip J. Rane, regional sales manager, SE Asia, OK International-India, points out, "India is poised at the threshold, as even with this growth rate the projected demand-supply gap is increasing, inviting more manufacturing in the electronics sector." But Chandramohan believes that as manufacturing starts increasing in

India, the market will start growing rapidly by 50 per cent. "It also depends on the government initiatives to restrict Chinese imports and encourage domestic manufacturing," he says.

Technological advancements

Talking about the promising potential of SMT market, the sector is also witnessing major technological innovations. Growing sophistication and shrinking size of electronic gadgets is fuelling these advancements. There is a recent demand in the industry to produce high-precision complex boards and assemblies. At the same time, there is a constant pressure to increase the quality of the goods produced, aiming for zero defects. To be able to manage this, manufacturers need to have the right SMT equipment, which will enable fast, flexible and perfect production of PCBs.

Nowadays, a wide range of machines are available with throughput as much as 25,000-45,000 components per hour (CPH) and mount-precision as good as 30 micron. High-speed changeover facilities, intelligent tape feeders and conveyors, 5-megapixel inline inspection systems, high-speed and high-resolution automatic optical inspection (AOI) systems, soldering robots, automatic high-speed printers with 12-second cycle-time and 10-micron repeatable accuracies at fine pitch, six-channel profiler packs for reflows, etc, are some of the examples of the latest technologies available today.

In order to cater to the demands coming from the manufacturers, SMT equipment are undergoing non-stop technical advancements. Some of the significant trends that are observed in the field are increased parallelisation and flexibility, high levels of software

India being a price sensitive country, the cost factor of SMT machines has been the most important hindrance in its market growth